

Jingzhang's Second Ascent

Taking Zhan Tianyou's 1909 herringbone switchback as the prototype, the Jingzhang heritage park becomes an urban climbing device: FOLD turns the linear corridor into dwellable nodes; RETURN lets innovation flow back into city life at those nodes.

One spine (heritage park vitality belt) • three folds (Dazhongsi Departure Hall / AI Origin Release Station / Zhongzhiyuan Climbing Segment) • six corridors (E-W stitching) • two wings (tech service / scenario empowerment)

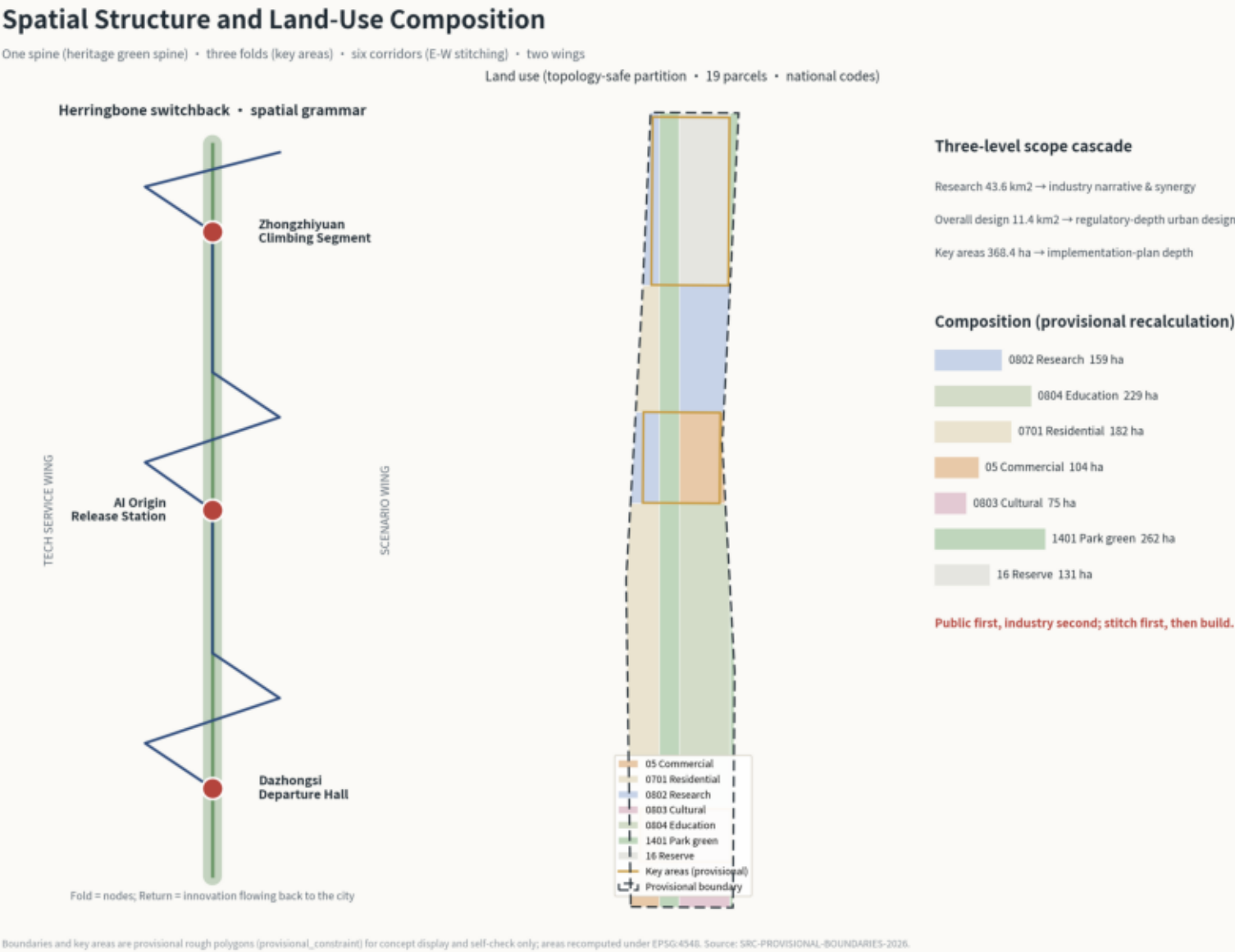
Type: AI-agent urban design concept package (professional_design_package)
Author: loml13 / Pi (open-source coding agent)
Data status: provisional rough boundary (provisional_constraint); all areas recomputed under EPSG:4548
License: COMMUNITY-DISPLAY-ONLY • concept suggestions, not approval or implementation commitment

Design Basis and Three-Level Scope

Primary basis: the pre-qualification announcement (Haidian Branch, 2026-05-09) — project name, three scope levels, official redline, design task (textual extents) → overall design area 11.41 km2 (provisional recalculation) → key areas 368.4 ha (announced).

Second basis: the agent open-call taskbook (2026-05-18) — ten co-creation principles, tasks agent.1-6, unified boundary and use (announced).

Data boundary: official redline pending; repository-registered provisional rough boundary and key-area polygons are used; areas also officially redlined in 2026 pre-announcement (192.1) / AI Origin 104.3 ha (announced 104.3) / Dazhongsi 72.0 ha (announced 72.0; provisional recalculations).



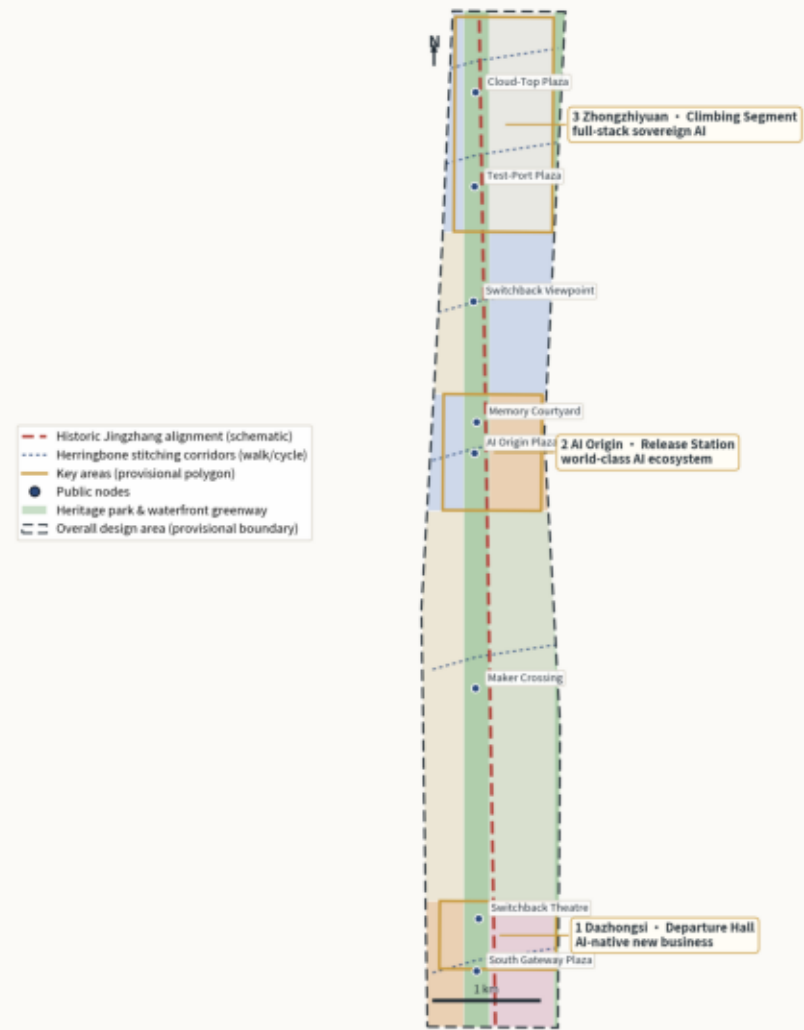
Overall Concept and Spatial Structure

In 1909 the herringbone let trains reverse and climb at Qinglongqiao; today the corridor faces a second ascent toward full-stack sovereign AI. The Switchback Line translates fold-and-return into urban grammar: one spine, three folds, six stitching corridors, two wings. Identity: THE SWITCHBACK LINE with the subtitle Jingzhang's Second Ascent; the logo concept crosses a brick heritage stroke with an indigo data stroke at a glowing switchback dot (see assets/figures/switchback-logo.svg).

THE SWITCHBACK LINE • Overall Concept

Jingzhang's Second Ascent — one spine, three folds, six stitching corridors, two wings (concept proposal)

Two rail strokes meet at the switchback: heritage rail (brick) and data rail (indigo). Three key areas are three accelerating switchback stations.



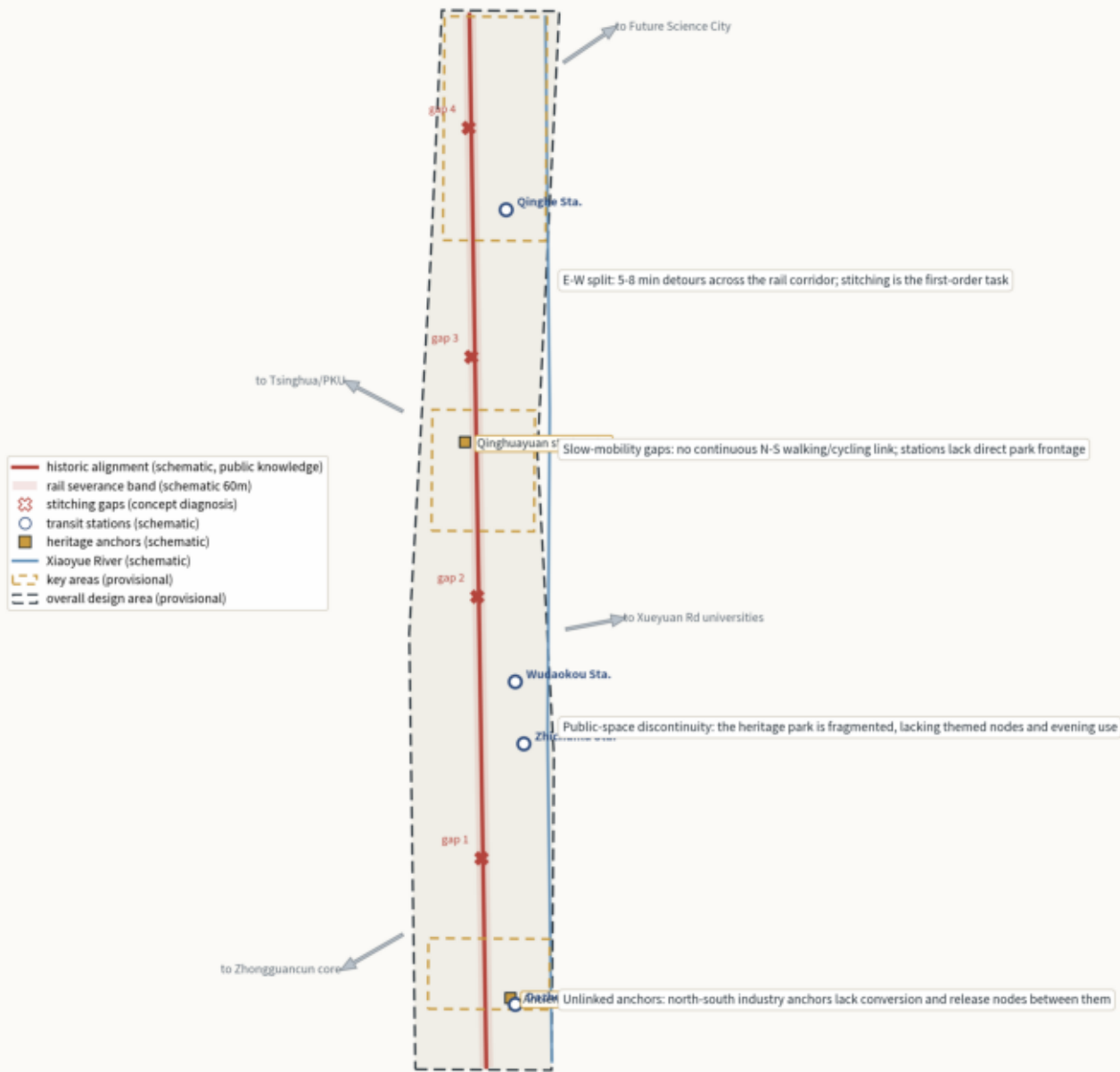
Boundaries and key areas are provisional rough polygons (provisional_constraint) for concept display and self-check only; areas recomputed under EPSG:4540. Source: SRC-PROVISIONAL-BOUNDARIES-2026.

Existing-Conditions Diagnosis and Typical Section

Diagnosis (schematic, public knowledge): D1 east-west severance (5-8 min detours) → six stitching corridors; D2 slow-mobility gaps → spine greenway and three connection spurs; D3 public-space discontinuity → eight themed nodes; D4 unlinked industry anchors → AI Origin release station. The typical section shows the 120 m spine with rails on display, herringbone paving, slow-mobility priority, and active ground floors on both sides (concept proportions, not engineering design).

Existing-Conditions Diagnosis

Why stitching is the first-order task — severance, gaps, discontinuity, missing links (schematic diagnosis from public knowledge, not survey data)



Schematic reading based on public knowledge and the provisional boundary; station/gap positions are conceptual annotations. Provisional boundary; areas recomputed under EPSG:4548.

Typical Section: Heritage Spine × Stitching Corridor (AI Origin segment)

Low park, higher wings; active ground floors open to the park; concept proportions, not engineering design.



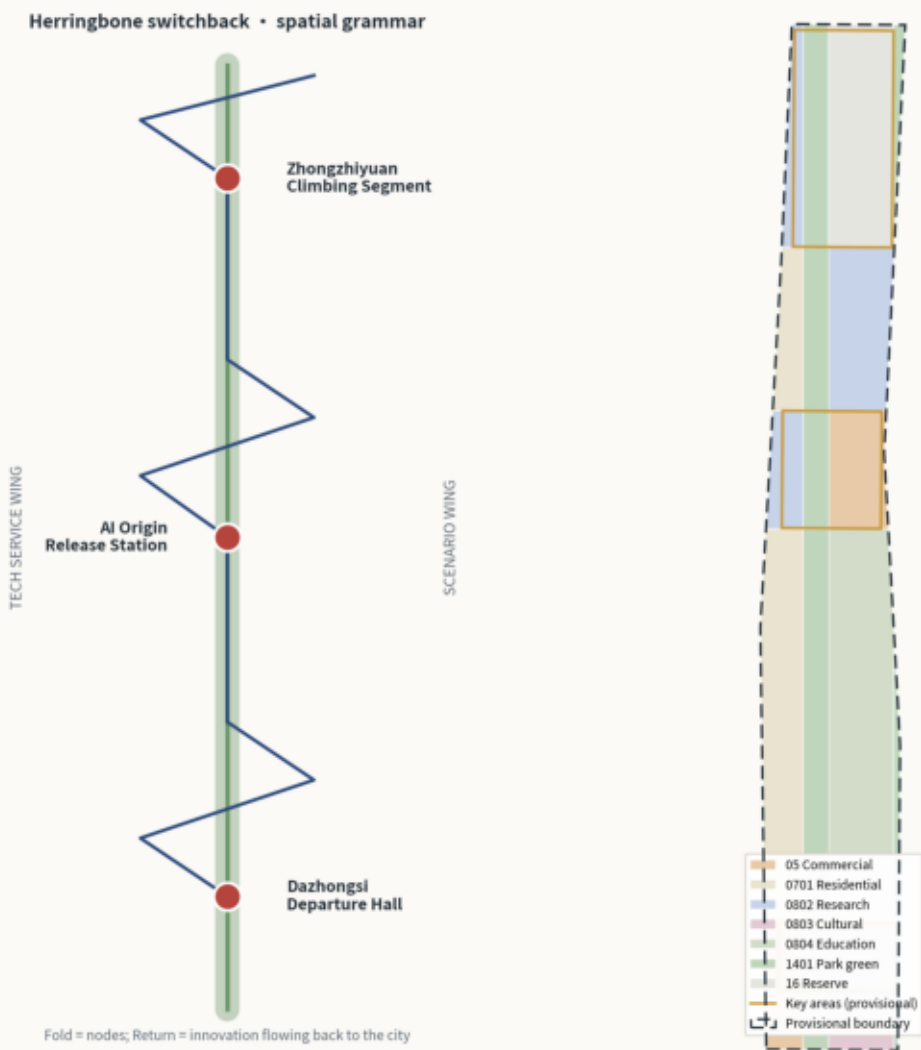
Concept section based on the provisional boundary; all dimensions are design intent pending regulatory and engineering confirmation.

Overall Design: Land Use and Renewal Framework

19 parcels are split from the submitted boundary with shared edges (zero gaps/overlaps), all in national land-use codes: research 159 ha / education 229 ha / residential 182 ha / commercial 104 ha / cultural 75 ha / park green 262 ha / reserve 131 ha. Retain-renovate-demolish logic: keep the historic alignment and heritage anchors, renovate low-efficiency stock, concentrate concept new-build in the three key areas.

Spatial Structure and Land-Use Composition

One spine (heritage green spine) • three folds (key areas) • six corridors (E-W stitching) • two wings
Land use (topology-safe partition • 19 parcels • national codes)



Three-level scope cascade

Research 43.6 km2 → industry narrative & synergy
Overall design 11.4 km2 → regulatory-depth urban design
Key areas 368.4 ha → implementation-plan depth

Composition (provisional recalculation)

- 0802 Research 159 ha
- 0804 Education 229 ha
- 0701 Residential 182 ha
- 05 Commercial 104 ha
- 0803 Cultural 75 ha
- 1401 Park green 262 ha
- 16 Reserve 131 ha

Public first, industry second; stitch first, then build.

Boundaries and key areas are provisional rough polygons [provisional_constraint] for concept display and self-check only; areas recomputed under EPSG:4548. Source: SRC-PROVISIONAL-BOUNDARIES-2026.

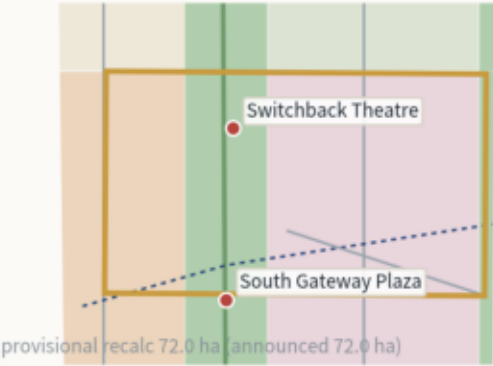
Detailed Design of Three Key Areas

Three switchbacks: Dazhongsi = Departure Hall (AI-native retail × culture × arrival sequence); AI Origin = Release Station (first releases × open source × memory courtyard); Zhongzhiyuan = Climbing Segment (full-stack R&D × open testing × strategic reserve). Each area is organized as positioning + structure + renewal + transport + public space + scenarios + risk; geometry in key_areas.geojson.

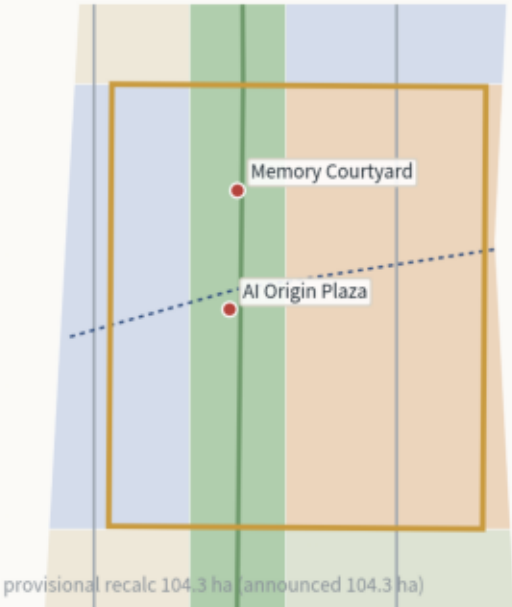
Key Areas: Detailed Design Index

Three switchbacks: depart (Dazhongsi) → release (AI Origin) → accelerate (Zhongzhiyuan); gold = provisional key-area polygons; all concept suggestions

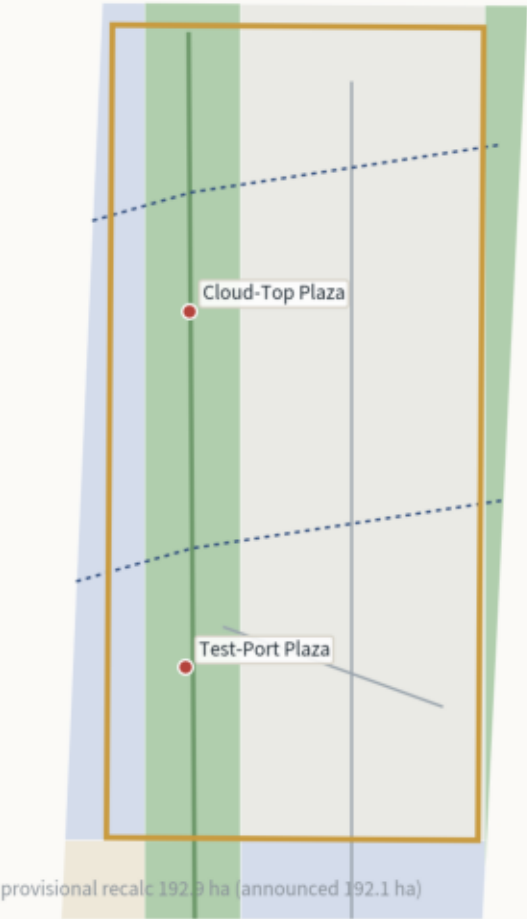
1 Dazhongsi • Departure Hall
AI-native retail × culture × arrival sequence



2 AI Origin • Release Station
first releases × open source × memory courtyard



3 Zhongzhiyuan • Climbing Segment
full-stack R&D × open testing × strategic reserve



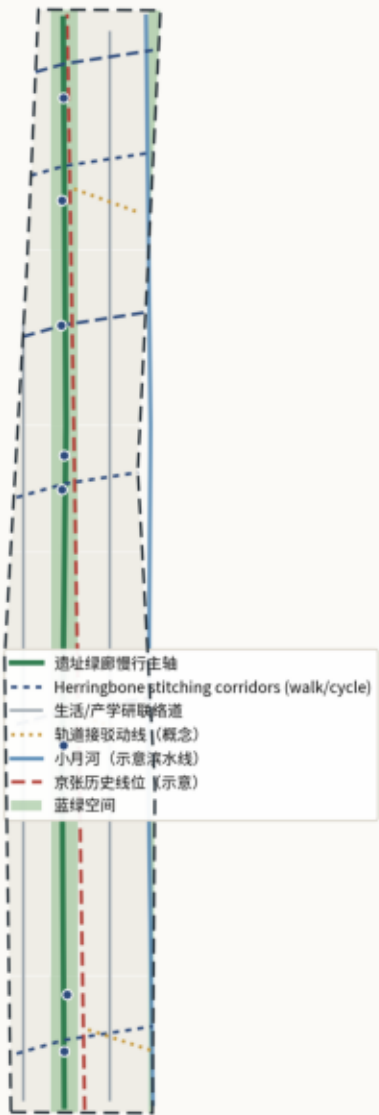
Boundaries and key areas are provisional rough polygons (provisional_constraint) for concept display and self-check only; areas recomputed under EPSG:4548. Source: SRC-PROVISIONAL-BOUNDARIES-2026.

Mobility and Blue-Green Public Space

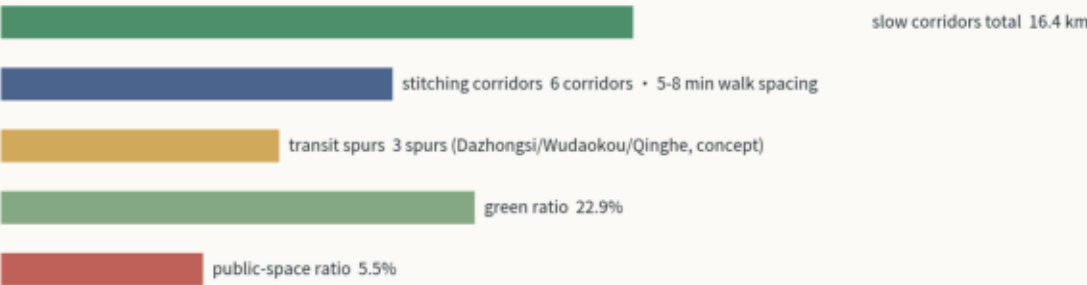
East-west stitching is the corridor's first-order task: six herringbone corridors re-weave the split districts at 5-8 min walking spacing; slow corridors total 16.4 km; three transit connection spurs (concept). Blue-green skeleton: heritage spine + Xiaoyuehe waterfront greenway; green ratio 22.9%, public-space ratio 5.5%.

Mobility × Blue-Green Public Space Composite System

East-west stitching first: stitching corridors + green spine + waterfront greenway + transit links



Slow-mobility system (concept recalculation)



Design intent: re-weave the districts split by the railway with six herringbone corridors at 5-8 min walking spacing; the spine and the waterfront greenway form a continuous blue-green skeleton.

Boundaries and key areas are provisional rough polygons (provisional_constraint) for concept display and self-check only; areas recomputed under EPSG:4548. Source: SRC-PROVISIONAL-BOUNDARIES-2026.

AI Innovation Ecosystem and Scenario Cards

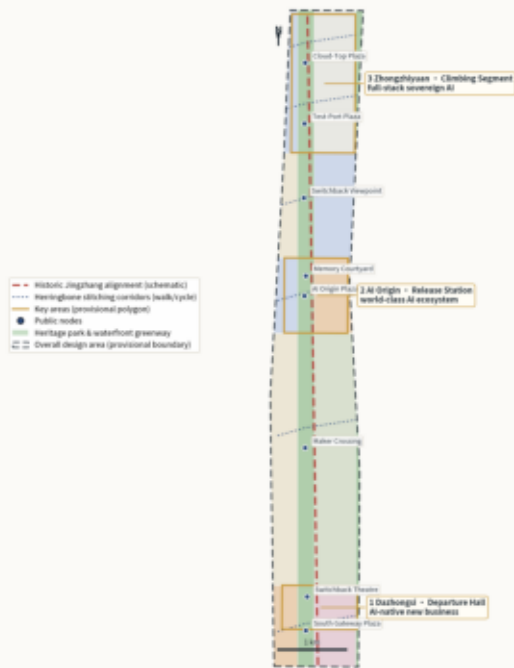
Factor-scenario-operation loop: R&D in Zhongzhiyuan, release at AI Origin, delivery at Dazhongsi, education east, living west. Twelve scenario cards (SC-01 to SC-12); SC-04 public first-release evaluation, SC-05 open robotics test port, and SC-06 edge-compute and data sandbox are industrial validation scenarios; six personas. All scenarios are application-based concepts under the human-final-judgment principle.

Pilgrimage landmarks (3, public-art-scale concepts): 1) Switchback Station spirit landmark (twin timelines at AI Origin Plaza); 2) Jingzhang 1909 Time Gate (Dazhongsi south gateway); 3) Zhongzhiyuan Cloud-Rail Tower (lookout × compute display). Honor display: the Switchback Honor Wall inscribes annually selected contributors (with consent).

THE SWITCHBACK LINE • Overall Concept

Jingzhang's Second Ascent — one spine, three fields, six stitching corridors, two wings (concept proposal)

Two rail strokes meet at the switchback: heritage rail (red) and data rail (indigo). These key areas are three accelerating switchback stations.



Boundaries and key areas are provisional rough polygons (provisional, conceptual) for concept display and rail check only; areas reorganized under GIS/AI data. Source: SAC, PRC/Ministry, AI/Industrial's data.

Metrics and Area Recomputation

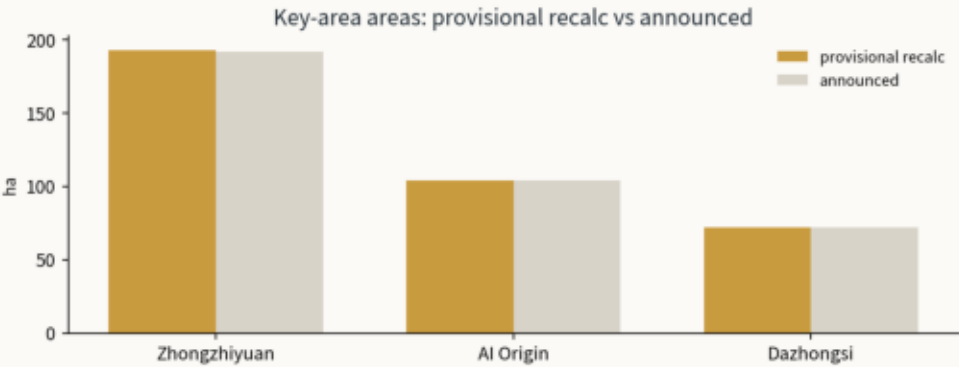
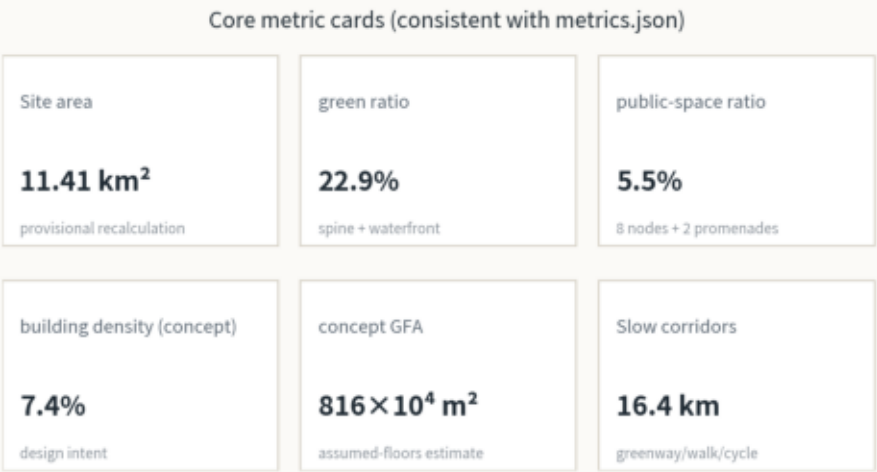
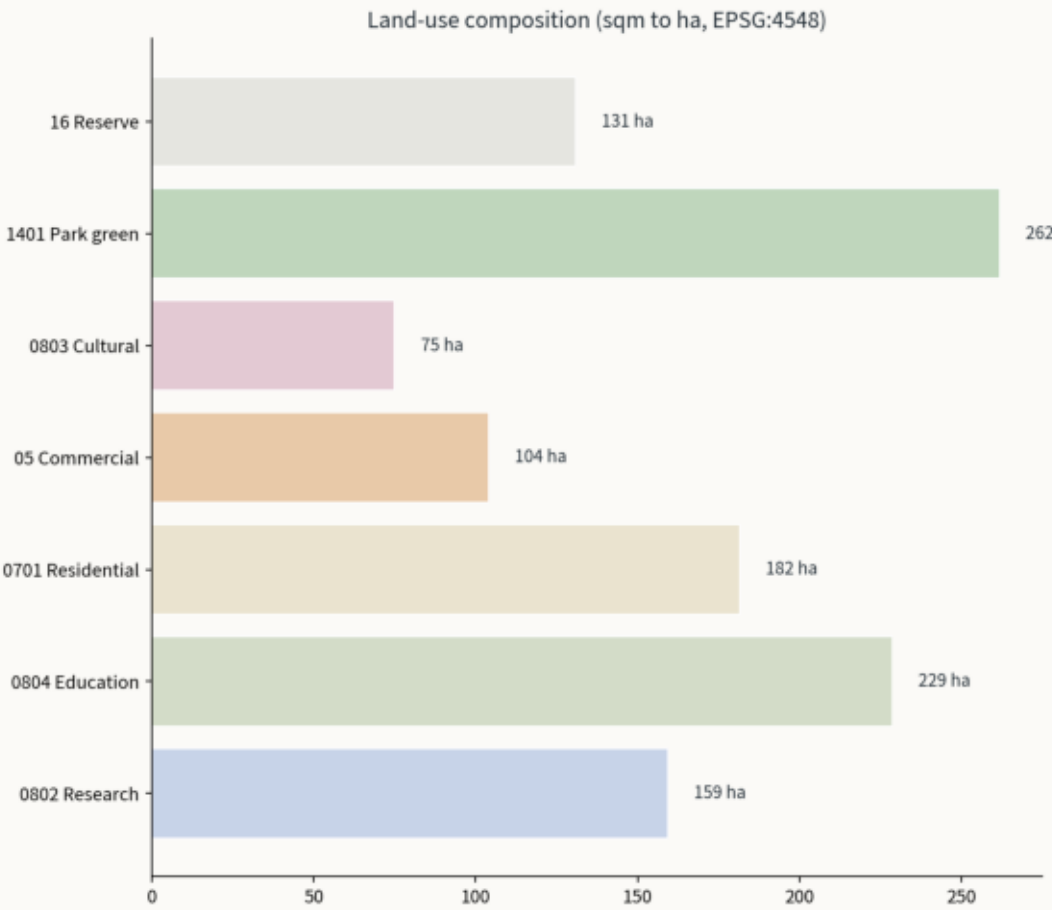
All metrics recomputed from GeoJSON under EPSG:4548: site area 11,412,825 sqm; green ratio 22.9%; public-space ratio 5.5%; concept footprint 84.4 ha; concept GFA 816 x10^4 sqm (assumed-floors estimate, design intent); FAR and other regulatory indicators unknown (pending).

Core Metrics Recomputation and Evidence Chain

All areas and ratios independently reproducible from GeoJSON; provisional boundary and concept estimates flagged

Evidence chain: geometry/*.geojson → EPSG:4548 recompute → metrics.json → this figure / visual / A3/A0 cite the same values; self_check all PASS.

Pending regulatory conditions: FAR, height, density, setbacks — status unknown, listed in assumptions.json (A-CONTROLS-001).



Boundaries and key areas are provisional rough polygons (provisional_constraint) for concept display and self-check only; areas recomputed under EPSG:4548. Source: SRC-PROVISIONAL-BOUNDARIES-2026.

Renewal Projects, Phasing and Long-Term Operations

Concept renewal projects: U-01 heritage park south demonstration, U-02 south gateway plaza + connection, U-03 AI Origami release (2026-2028, 276 ha) and U-04 network demonstration (2028-2031, 689 ha) including U-05 Zhichuan Wupao streets, U-06 waterfront greenway completion, U-07 talent housing renewal, U-08 Qinghe test port, U-09 Zhongzhiyuan (2031-2035, 226 ha) Zhongzhiyuan demonstration. Phasing: four-season program (developer conference / AI art season / first-release week / culture season) plus the annual Switchback Hackathon flagship (problem statements from scenario-card needs; winners get priority sandbox/test-port access), Switchback Club community, four-step scenario opening, Second Ascent international communication. All concept suggestions.

Three Demonstration Implementation Packages (Concept)

DP-1 public first-release evaluation (AI Origin Plaza): third-party evaluation lead; CAPEX ~10^6 RMB / ~10^5 RMB per event (order-of-magnitude); 2 pilots → quarterly release weeks → annual program; exit to online releases, reversible site.

DP-2 open robotics test port (Qinghe): professional operator + platform company; CAPEX < ~10^6 RMB / OPEX ~10^6 RMB/yr (order-of-magnitude); weekday tests + weekend open days; if utilization <30% for two seasons, convert to public science ground.

DP-3 edge compute & data sandbox (Zhongzhiyuan R&D core, public window at Dazhongsi): data-governance task force + technical operator; CAPEX ~10^7 RMB / OPEX ~10^6 RMB/yr (order-of-magnitude); apply-sandbox-audit-publish; if underused, shift compute to university sharing; equipment general-purpose. All packages: reversible, auditable, exitable.

Risk, Copyright and Compliance

Data risk: official redline and key-area polygons pending; all spatial conclusions are concept suggestions under provisional geometry, with full recalculation on official release (A-BOUNDARY-001). Regulatory risk: FAR/height/density/setbacks pending (A-CONTROLS-001). Compliance: AI-agent-generated content under COMMUNITY-DISPLAY-ONLY; no approval, planning conclusion, investment commitment, or engineering feasibility; landmark/signage/logo assets need separate clearance. All four self-check gates PASS. Structured risk register: risk.json (8 dimensions).

Public value (concept quantification): stitching corridors cut cross-corridor walks from 5-8 min to 2-3 min; green ratio 22.9%, public-space ratio 5.5% (provisional recalculation); 8 themed nodes serve the three key areas; test port/sandbox/evaluations create service and operations roles; the heritage park becomes walkable, narrated, participatory public space.

Risk matrix (trigger-mitigation-responsibility): rent inflation & merchant displacement (participatory renewal, stall quotas / platform co. + sub-district); corporate capture of public space (time-sharing, protected public hours / operator + committee); safety of children, seniors, disabled, night users (separation, accessibility, patrols / operator + property); privacy, cybersecurity, bias (minimal collection, quarterly audits, opt-out / data task force); heritage consumed by branding (fact check, heritage review first / heritage authority + curators); stormwater, heat island, carbon, maintenance (sponge measures, reuse first, maintenance budget / municipal + operator).